What Does the Price of TIPS Depend on?

Paper Presentation

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10 June 2026

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Scope and Motivation

This paper studies the determinants of the yield spread between a nominal US Treasury and its replicating portfolio (TIPS and Inflation Swap), providing a model-free decomposition of expected real rates, inflation and associated risk premia.

Why and what is our contribution?

1 Motivation

- TIPS are central instruments for extracting breakeven inflation rates (BEI).
- Evidence of market arbitrage if BEI $\neq$ ISR as per Fleckenstein et al. (2014).
- Vast literature on TIPS liquidity premia (e.g. Abrahams et al. (2016) and D'Amico et al. (2018)), **little focus on inflation surprises**.
- Understanding how inflation expectations behaved during the 2022 inflation surge.

2 Contributions

- Two goals: (i) assess effects of **liquidity and inflation surprises** on TIPS yields, and (ii) decompose **nominal and real yields into expectations and risk premia**.
- Proxies for liquidity: VIX, MOVE, PCA.
- Measures of expected inflation and nominal yields in $\mathbb{P}$ (*Blue Dots*).

Our Results at a Glance

Our findings show that: (i) TIPS yields offer a liquidity premium relative to US Treasury bonds and depend on inflation surprises, (ii) inflation expectations remained anchored to the FED target following the post-COVID surge in inflation dynamics (2022), and (iii) real risk premia move inversely to inflation expectations as TIPS represent a hedging tool during inflationary conditions.

- The price discrepancy between a nominal Treasury bond and its replicating portfolio (long TIPS and short ISR) depends not only on liquidity **but also on inflation surprises**. This is at first blush surprising, because, apart from minor effects, the price of TIPS is derived by discounting its real cashflows at the real rate.

- TIPS are less liquid than nominal Treasuries. Since states of poor market liquidity are associated with phases of high market distress, TIPS yields offer a liquidity premium.

- Although TIPS yields do not depend on expected inflation, we observe the **demand for TIPS to increase during episodes of positive inflation surprises**, pushing up TIPS prices (and compressing yields).

- Studies conducted before the 2022 inflation shock couldn't disentangle these two effects since the greatest liquidity shock was observed during GFC, where inflation expectations and liquidity pulled TIPS yields in the same direction.

- The inflation risk premium has increased during phases of rising inflation, remaining, however, in line with historical timeseries.

Literature Review

We frame the literature on TIPS pricing by focusing on why breakeven inflation differs from the corresponding inflation swap rate. Hence, our focus is split into three different pieces of literature: (i) TIPS mispricing, (ii) TIPS Liquidity Premia and 3) Research Tools used throughout the article.

TIPS Mispricing

- Fleckenstein et al. (2014): arbitrage among UST, ISR and TIPS.
- Bretscher (2015)
- To and Tran (2019)

TIPS Liquidity Premia

- Haubrich et al. (2012)
- Abrahams et al. (2016)
- D'Amico et al. (2018): liquidity as the missing factor.

Research Tools

- Gürkaynak et al. (2007) and Gürkaynak et al. (2010): UST and TIPS ZCB yields.
- Drechsler et al. (2020): VIX and MOVE for liquidity risk.
- Rebonato and Sherwin (2020): PCA for liquidity measure.
- Rebonato and Ronzani (2020): DMRV for expected nominal yields in $\mathbb{P}$.

Setup: Data Description

This paper employs zero-coupon yields/rates for nominal Treasuries, TIPS and inflation swaps, assuming monthly time-series starting in August 2007 up to September 2024. All data are publicly available on Bloomberg, the Bureau of Labor Statistics and the FED websites.

Description

- Zero-coupon yields/rates for bonds/ISR **(e.g. no CPI seasonality or deflation floor)**.
- Monthly timeseries aligned on CPI publication dates **(no month-end)**.
- Tenor Range (T): 2yrs up to 20yrs.
- Time Horizon (t): August 2007 up to September 2024 **(timeseries restrictions/ more established TIPS market)**.

Timeseries

- UST and TIPS: FED estimates **(fitted curve, no bond level data)**.
- Liquidity Proxies: VIX, MOVE **(direct)**, on-/off-the-run 10y UST spread, *HPW* index **(PCA)**.
- Expected Inflation: actual SA CPI and FED estimates.
- Expected Nominal Yields: *Dot Plot* (quarterly).

The data have two dimensions **(timeseries and cross-sectional)** and cover several structural changes (e.g. GFC, COVID-19 pandemic and the subsequent inflation surge). This allows us validating whether **certain tenors are more sensitive to a specific factor** (e.g. liquidity).

Setup: Assumptions and Definitions

We rely on three key assumptions to implement the yield decomposition of UST (Y_t^T), TIPS (y_t^T) and ISR (ISR_t^T). Yields are defined as the sum of expectations and corresponding risk premia.

Assumptions

1. λ_t^T as ***latent*** factor impacting TIPS yields (y_t^T). Liquidity proxy?
2. Inflation swap rate curve (ISR_t^T) not affected by liquidity **(mid price unchanged)**. To be validated.
3. Negligible yield convexity (e.g. Rebonato et al. (2020)).

Definitions

- UST yields: $Y_t^T = \hat{r}_t^T + \hat{\pi}_t^T + rpr_t^T + rp\pi_t^T$
- TIPS yields: $y_t^T = \hat{r}_t^T + rpr_t^T + \lambda_{\mathbf{t}}^{\mathbf{T}}$
- Inflation swap rates: $ISR_t^T = \hat{\pi}_t^T + rp\pi_t^T$
- Breakeven Inflation: $BEI_t^T = Y_t^T - y_t^T = \hat{\pi}_t^T + rp\pi_t^T - \lambda_{\mathbf{t}}^{\mathbf{T}}$

Methodology: Overview

Two goals, hence two methods. The first identifies whether λ_t^T reflects liquidity and inflation surprises via OLS regressions. The second decomposes yields into risk premia and expectations using inflation and nominal-rate expectations.

Factors Affecting TIPS Yields

Goal: λ_t^T ?

- Exposure to market **liquidity**
- Exposure to **inflation surprises**

→

Method

- OLS regressions (by T) applied to λ_t^T on separate inputs
- Second assumption validation: **(ISR not impacted by liquidity)**

→

Inputs

- **Two measures for λ_t^T**: $ISR_t^T - BEI_t^T$ and $\hat{\gamma}_t^T = BEI_t^T - \hat{\pi}_t^T = rp\hat{\pi}_t^T - \lambda_t^T$
- Liquidity Proxies: VIX **(observable)** and PCA **(indirect)**
- Inflation Surprise: YoY Inflation - 10yr Moving Average

Decomposition of Nominal and Real Yields

Goal: Decomposition

- Extract expectations and risk premia for real rate and inflation

→

Method

- Linear system of equations for $Y_t^T, y_t^T, \hat{\gamma}_t^T$ and ISR_t^T
- **Four observable** quantities but **five unknown** variables

→

Inputs

- **Observable**: $Y_t^T, y_t^T, \hat{\gamma}_t^T$ and ISR_t^T
- **Estimated**: $\hat{r}_t^T$ and $\hat{\pi}_t^T$

Methodology: Factors Affecting TIPS Yields

We separately regress λ_t^T, ISR_t^T, $\hat{\gamma}_t^T$ on liquidity, and λ_t^T on a proxy for inflation surprises to assess whether λ_t^T is a premium for liquidity. These results have been tested in a SUR framework as well without material changes.

OLS Regressions

1. $\hat{\gamma}_t^T = \alpha_{\hat{\gamma}}^T + \beta_{\hat{\gamma}}^T PCA_t + \epsilon_{t,\hat{\gamma}}^T$
2. $\lambda_t^T = \alpha_{\lambda}^T + \beta_{\lambda}^T PCA_t + \epsilon_{t,\lambda}^T$
3. $ISR_t^T = \alpha_{ISR}^T + \beta_{ISR}^T PCA_t + \epsilon_{t,ISR}^T$
4. $\lambda_t^T = \alpha_{InfS}^T + \beta_{InfS}^T InfS_t + \epsilon_{t,InfS}^T$

Key Observations

- Regressions **(1)** and **(2)** to identify whether λ_t^T is affected by liquidity.
- Limited/No dependence of ISR_t^T on liquidity in **(3)** to confirm our second assumption.
- The separate regressions of λ_t^T on liquidity and inflation surprises in **(2)** and **(4)** estimate impact across T.
- SUR for same-time cross-correlation of residuals (same regressors, no change in coefficients).

Methodology: Liquidity Proxies

Liquidity is not directly observable, hence we make use of two proxies: VIX (direct) as per Drechsler et al. (2020) and PCA (estimated) in the sense of Rebonato and Sherwin (2020). Since VIX may capture more than just liquidity, PCA is our preferred measure.

Measures of Liquidity

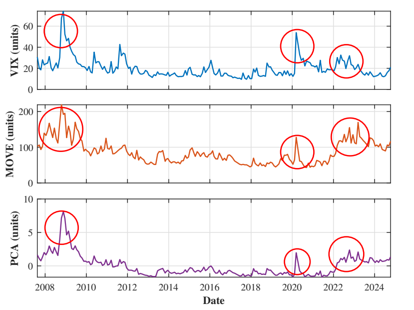
- **VIX**: liquidity risk is akin to short exposure on volatility risk.
- **PCA**: first principal component extracted from VIX, MOVE, on-/off-the-run 10y UST spread, and *HPW* index.
- Each variable is standardized and demeaned.

$\downarrow$ **PCA**

Relevance $\Rightarrow$ 64% of variance explained

Correlation $\Rightarrow \rho >$ 80% vs VIX, MOVE, and *HPW*

Timeseries: VIX, MOVE and PCA



Methodology: Yield Decomposition

We focus now on the decomposition of nominal and real yields into risk premia and expectations. We have a linear system binding y_t^T, Y_t^T and ISR_t^T and close it with estimates for expected inflation and nominal yields.

Linear System

$$\begin{cases} Y_t^T = \hat{\pi}_t^T + \hat{r}_t^T + rpr_t^T + rp\pi_t^T \\ y_t^T = \hat{r}_t^T + rpr_t^T + \lambda_t^T \\ \hat{\gamma}_t^T = rp\pi_t^T - \lambda_t^T \\ ISR_t^T = \hat{\pi}_t^T + rp\pi_t^T \end{cases}$$

Expected Nominal Yields: $\hat{R}_t^T$

- Doubly-Mean-Reverting Vasicek **(Rebonato and Ronzani (2020))**
- Applied to FED *Blue Dots* to get $\hat{R}_t^T$
- $\hat{r}_t^T = \hat{R}_t^T - \hat{\pi}_t^T$

Expected Inflation: $\hat{\pi}_t^T$

- Two measures: FED **(Haubrich et al. (2012))** and internal
- AR(1) on 12m trailing inflation rate **(SA CPI)**
- **How do they compare?**

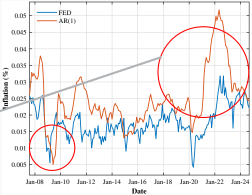
Methodology: Expected Inflation

Expected inflation is not directly observable, hence must be estimated. The two chosen metrics: FED and AR(1) have similar trends and are consistent in long-term expectations. We expect our results to be robust across both specifications.

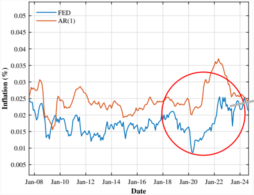
Expected Inflation: AR(1) vs FED

AR(1) more volatile

Expected Inflation: 2-year



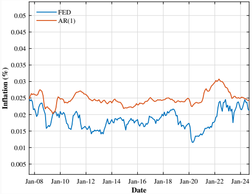
Expected Inflation: 5-year



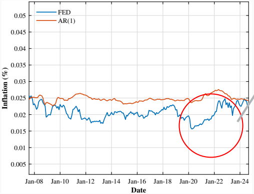
FED quicker convergence

Higher reversion speed

Expected Inflation: 10-year



Expected Inflation: 20-year

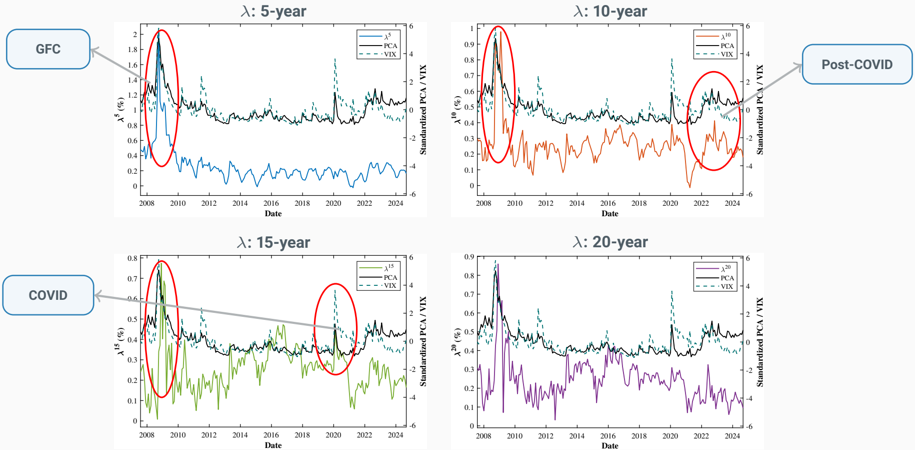


Long-Term $\Rightarrow 2 - 2.5\%$

FED more volatile

In line with the first goal of this paper, we focus on λ and whether it depends on liquidity and inflation surprises. λ and liquidity (VIX and PCA) appear to co-move over time (e.g. GFC or COVID-19).

λ vs VIX and PCA



Results and Discussion: λ

There is a consistent positive relationship between λ and liquidity as suggested by regression statistics. Nevertheless, size of the coefficients and explained variance fade away as TIPS tenor increases. Subject to further research, this phenomenon might be explained by investor segmentation...

Key Considerations

- $\hat{\beta}$: generally positive, decreasing against TIPS tenor.
- R^2: meaningful explained variance up to 10yrs tenor.
- Similar results obtained for VIX.

Economic Relevance $\Rightarrow$ short/medium term

Inflation Surprise $\Rightarrow$ missing piece?

λ and PCA: Regression Results

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{PCA}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{PCA}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\lambda}^{2Y}$	0.2464	0.1053	0.0126	0.0079	19.549	0.0000	13.349	0.0000	0.466
$\hat{\lambda}^{3Y}$	0.2580	0.1299	0.0107	0.0067	24.018	0.0000	19.311	0.0000	0.646
$\hat{\lambda}^{4Y}$	0.2594	0.1352	0.0100	0.0063	25.918	0.0000	21.574	0.0000	0.695
$\hat{\lambda}^{5Y}$	0.2566	0.1232	0.0088	0.0055	29.126	0.0000	22.331	0.0000	0.710
$\hat{\lambda}^{6Y}$	0.2514	0.1032	0.0079	0.0049	31.949	0.0000	20.949	0.0000	0.683
$\hat{\lambda}^{7Y}$	0.2502	0.0904	0.0075	0.0047	33.414	0.0000	19.290	0.0000	0.646
$\hat{\lambda}^{8Y}$	0.2463	0.0710	0.0071	0.0044	34.725	0.0000	15.976	0.0000	0.556
$\hat{\lambda}^{9Y}$	0.2478	0.0556	0.0070	0.0044	35.192	0.0000	12.608	0.0000	0.438
$\hat{\lambda}^{10Y}$	0.2556	0.0439	0.0074	0.0046	34.626	0.0000	9.503	0.0000	0.307
$\hat{\lambda}^{12Y}$	0.2533	0.0142	0.0074	0.0046	34.161	0.0000	3.065	0.0025	0.044
$\hat{\lambda}^{20Y}$	0.2403	0.0158	0.0076	0.0048	31.665	0.0000	3.316	0.0011	0.051

Results and Discussion: λ

...Inflation Surprise is the variable that explains long-term λ. On one side, long-dated TIPS are mainly held by strategic players (e.g. lifers) which don't react on liquidity changes but on inflation shocks, by bidding TIPS in an inflationary environment hence compressing λ. On the other hand, short-dated TIPS are held by traders (e.g. banks) which are sensitive to market liquidity.

Key Observations

- $\hat{\beta}$: generally negative, inflation shocks compress λ.
- R^2: meaningful at long tenors.
- **Economic Relevance** $\Rightarrow$ long-term. **Why?**

Two effects

Market liquidity $\Rightarrow$ short/medium term

Bidding pressure $\Rightarrow$ long-term inflation shocks

λ and InfS: Regression Results

Tenor	$\hat{\alpha}$	$\hat{\beta}_{InfS}$	SE($\hat{\alpha}$)	SE($\hat{\beta}_{InfS}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\lambda}^{2Y}$	0.2737	-1.1211	0.0272	0.8385	10.057	0.0000	-1.337	0.1827	0.009
$\hat{\lambda}^{3Y}$	0.2862	-1.1678	0.0285	0.8780	10.045	0.0000	-1.330	0.1850	0.009
$\hat{\lambda}^{4Y}$	0.2843	-1.0350	0.0286	0.8821	9.930	0.0000	-1.173	0.2420	0.007
$\hat{\lambda}^{5Y}$	0.2717	-0.6400	0.0259	0.7970	10.503	0.0000	-0.803	0.4229	0.003
$\hat{\lambda}^{6Y}$	0.2668	-0.6487	0.0221	0.6804	12.084	0.0000	-0.953	0.3415	0.004
$\hat{\lambda}^{7Y}$	0.2656	-0.6442	0.0199	0.6125	13.360	0.0000	-1.052	0.2942	0.005
$\hat{\lambda}^{8Y}$	0.2645	-0.7465	0.0168	0.5169	15.765	0.0000	-1.444	0.1502	0.010
$\hat{\lambda}^{9Y}$	0.2687	-0.8509	0.0148	0.4545	18.212	0.0000	-1.872	0.0626	0.017
$\hat{\lambda}^{10Y}$	0.2843	-1.1575	0.0138	0.4251	20.605	0.0000	-2.723	0.0070	0.035
$\hat{\lambda}^{12Y}$	0.2883	-1.3991	0.0116	0.3569	24.889	0.0000	-3.920	0.0001	0.071
$\hat{\lambda}^{20Y}$	0.3016	-2.4476	0.0110	0.3392	27.400	0.0000	-7.216	0.0000	0.205

Results and Discussion: *ISR/Liquidity* Check

The identification of λ lies on the assumption that ISR do not embed compensation for liquidity risk, therefore any premium applied by dealers on ISR levels is reflected into the bid-ask spread leaving the mid unbiased: $ISR_t^T = \hat{\pi}_t^T + rp\pi_t^T$. This is tested by regressing ISR_t^T on liquidity proxies.

ISR and PCA: Regression Results

Tenor	$\hat{\alpha}$	$\hat{\beta}_{PCA}$	$SE(\hat{\alpha})$	$SE(\hat{\beta}_{PCA})$	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
ISR^2	1.8515	-0.2107	0.0579	0.0362	31.987	0.0000	-5.814	0.0000	0.148
ISR^3	1.9548	-0.1338	0.0465	0.0291	42.069	0.0000	-4.600	0.0000	0.095
ISR^4	2.0361	-0.0854	0.0395	0.0247	51.543	0.0000	-3.454	0.0007	0.056
ISR^5	2.1041	-0.0575	0.0353	0.0221	59.682	0.0000	-2.603	0.0099	0.032
ISR^6	2.1603	-0.0398	0.0322	0.0202	67.010	0.0000	-1.972	0.0500	0.019
ISR^7	2.2112	-0.0179	0.0296	0.0185	74.716	0.0000	-0.968	0.3341	0.005
ISR^8	2.2508	-0.0066	0.0279	0.0175	80.588	0.0000	-0.376	0.7070	0.001
ISR^9	2.2878	0.0045	0.0265	0.0166	86.440	0.0000	0.270	0.7877	0.000
ISR^{10}	2.3243	0.0146	0.0250	0.0157	92.919	0.0000	0.932	0.3524	0.004
ISR^{12}	2.3626	0.0151	0.0242	0.0152	97.462	0.0000	0.992	0.3225	0.005
ISR^{15}	2.3927	0.0187	0.0237	0.0148	101.060	0.0000	1.264	0.2076	0.008
ISR^{20}	2.4113	0.0238	0.0240	0.0150	100.360	0.0000	1.580	0.1157	0.012

Key Points

- $\hat{\beta}$: hovering around zero.
- R^2: not meaningful across term.
- **Economic Relevance** $\Rightarrow$ some, only at the very short-term (noisier points).
- Same applies for VIX.

$\downarrow$

Dependence on Liquidity $\Rightarrow$ can be disregarded

Results and Discussion: *ISR/Liquidity* Check

If λ captured the liquidity component embedded in the BEI, then, since $\hat{\gamma}_t^T = rp\pi_t^T - \lambda_t^T = BEI_t^T - \hat{\pi}_t^T$, it should reflect the same dependence on liquidity proxies assuming that the inflation risk premium does not exhibit a systematic dependence on liquidity. This is tested by regressing $\hat{\gamma}_t^T$ on liquidity proxies.

$\hat{\gamma}_t^T$ and PCA: Regression Results

Tenor	$\hat{\alpha}^T$	$\hat{\beta}^T_{PCA}$	SE($\hat{\alpha}^T$)	SE($\hat{\beta}^T_{PCA}$)	$t(\alpha)$	$p(\alpha)$	$t(\beta)$	$p(\beta)$	R^2
$\hat{\gamma}^{2Y}$	-0.1739	-0.4005	0.0487	0.0305	-3.573	0.0004	-13.141	0.0000	0.458
$\hat{\gamma}^{3Y}$	-0.0593	-0.3530	0.0401	0.0251	-1.481	0.1403	-14.068	0.0000	0.492
$\hat{\gamma}^{4Y}$	0.0219	-0.3111	0.0350	0.0219	0.624	0.5332	-14.180	0.0000	0.496
$\hat{\gamma}^{5Y}$	0.0835	-0.2710	0.0319	0.0200	2.617	0.0095	-13.563	0.0000	0.474
$\hat{\gamma}^{6Y}$	0.1301	-0.2326	0.0300	0.0188	4.339	0.0000	-12.385	0.0000	0.429
$\hat{\gamma}^{7Y}$	0.1643	-0.1968	0.0289	0.0181	5.688	0.0000	-10.880	0.0000	0.367
$\hat{\gamma}^{8Y}$	0.1880	-0.1646	0.0283	0.0177	6.640	0.0000	-9.286	0.0000	0.297
$\hat{\gamma}^{9Y}$	0.2030	-0.1368	0.0281	0.0176	7.228	0.0000	-7.782	0.0000	0.229
$\hat{\gamma}^{10Y}$	0.2106	-0.1136	0.0280	0.0175	7.516	0.0000	-6.476	0.0000	0.171
$\hat{\gamma}^{12Y}$	0.1487	-0.0766	0.0277	0.0174	5.366	0.0000	-4.418	0.0000	0.087
$\hat{\gamma}^{20Y}$	0.1035	-0.0628	0.0267	0.0167	3.883	0.0001	-3.761	0.0002	0.065

Key Points

- $\hat{\beta}$: negative, consistent with how λ_t^T impacts $\hat{\gamma}_t^T$.
- $\boldsymbol{R^2}$: similar to what observed for λ_t^T regressed on PCA.
- $\hat{\gamma}_t^T \Rightarrow$ difference in expected inflation under $\mathbb{Q}$ and $\mathbb{P}$.
- Results in line for $\hat{\gamma}_t^T$ VIX.

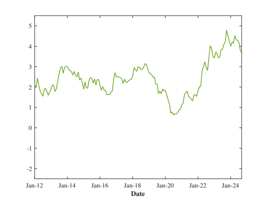
***ISR/Liquidity* Assumption** $\Rightarrow$ validated

Results and Discussion: Yield Decomposition

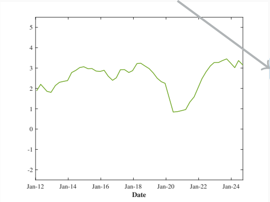
Now that we have answered the first question of this paper, we focus on the decomposition of nominal and real yields into risk premia and expectations. Results are presented for the 10-Year Tenor (similar conclusions for the other tenors).

System Identification: 10-Year Tenor

Nominal UST



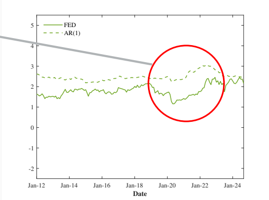
Expected Nominal Rate



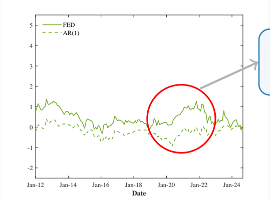
Blue Dots $\Rightarrow \hat{r}^{10} + \hat{\pi}^{10}$

Expected Inflation

FED vs AR(1)



Inflation Risk Premium



FED $\Rightarrow rp\pi^{10} > 0$

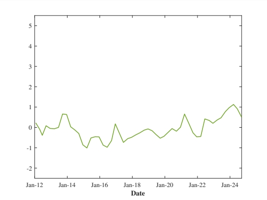
AR(1) $\Rightarrow rp\pi^{10} \sim 0$

Results and Discussion: Yield Decomposition

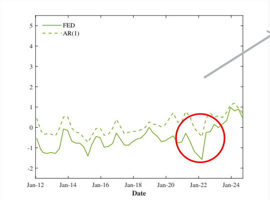
During phases of high inflation, investors sacrifice a portion of real risk premium to get inflation protection. Additionally, we see that FED expected inflation results in a higher inflation risk premium, even if structurally lower than AR(1).

System Identification: 10-Year Tenor

Nominal Risk Premium



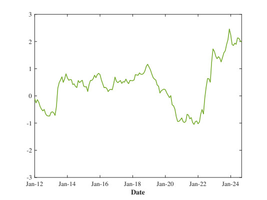
Real Risk Premium



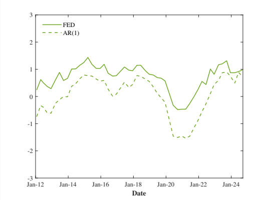
FED $\Rightarrow rpr^{10} < 0$

AR(1) $\Rightarrow rpr^{10} < 0$

TIPS Yield



Expected Real Rate



Conclusions

This paper finds that TIPS do not lend themselves to a market arbitrage, and investigates on the effects of market liquidity and inflation surprises on TIPS yields (and prices). Further analysis is provided for all attending measures due to the suggested yield decomposition, encompassing both the timeseries and cross-sectional dimensions of the data.

- TIPS yields depend on market liquidity and inflation surprises.
- While the first finding is a rather stylized fact: TIPS are less liquid than nominal Treasuries, the dependence of TIPS yields on **inflation surprises is quite original** since inflation should not matter.
- Due to the combination of timeseries and cross-sectional data, we show that TIPS yields are not impacted homogeneously by both factors (liquidity and inflation): market liquidity is mostly a driver for short/medium tenors while **inflation surprise becomes more meaningful for long-term TIPS.**
- We conjecture this is a result of the **investor segmentation of the TIPS market**: dealers at short/medium end and strategic for the long-term. Linking TIPS holdings to investor type is a suggestion for future studies.
- Inflation expectations (and risk premia) remained anchored, however FED estimates appear lower and converge more rapidly to the long-term target: this could reflect **FED's confidence that inflation is under control**.

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